

ECE 586IC: Interplay between Control and Machine Learning

Fall 2026

Instructor: Bin Hu (binhu7@illinois.edu)

Course Information:

Lectures: T/Th 12:00PM–1:20PM, Room 3015 ECEB

Office Hours: By appointment, CSL 145

Prerequisites: ECE 515; ECE 534 and ECE 490 are recommended, but not required.

Course Website: <https://binhu7.github.io/courses/ECE586IC/Fall2026/ECE586IChome.html>

Course Description:

This advanced graduate course focuses on the interplay between control and machine learning. One half of the course focuses on tailoring control tools to study algorithms and network models in large-scale machine learning. In the other half of the course, students will study how to combine machine learning and model-based control methods for control design problems.

Topics include empirical risk minimization; first-order methods for large-scale machine learning; stochastic optimization; dissipation inequalities; jump system theory; Lur’e–Postnikov type Lyapunov functions; integral quadratic constraints; the KYP lemma; graphical interpretations of optimization methods; control-theoretic methods for designing certifiably robust neural networks; control-oriented analysis of temporal difference learning and Q-learning; reinforcement learning for linear quadratic regulator (LQR) problems; zeroth-order optimization and evolutionary strategies; policy gradient methods for robust control; Goldstein’s subgradient method for H_∞ control; adversarial reinforcement learning; imitation learning for control; constrained policy optimization; diffusion models and generative AI for control; and large language models (LLMs) for control.

Course Outline:

- **Control-theoretic foundations for machine learning:** dissipativity, quadratic constraints, integral quadratic constraints, semidefinite programming, jump systems, and related Lyapunov methods.
- **Optimization and learning algorithms through a control lens:** empirical risk minimization, first-order methods, stochastic optimization, distributed optimization, and graphical/control-theoretic interpretations of optimization algorithms.
- **Learning dynamics and neural networks:** temporal difference learning, Q-learning, statistical learning for system identification and control, and verification of certifiably robust neural networks.
- **Reinforcement learning and optimization for control:** policy optimization for LQR and robust control, zeroth-order optimization, evolutionary strategies, Goldstein’s subgradient method for H_∞ control, and adversarial reinforcement learning.
- **Learning-based control:** imitation learning, constrained policy optimization, and safety of learning-enabled control systems.
- **Generative AI and foundation models for control:** diffusion models, generative policies, large language models, and foundation-model approaches for robotics and control.

Grading:

40% regular homework sets (4 homework sets in total, 10% each); 20% in-class presentation; 40% written research report.

Required Materials:

There is no required textbook for the class. All course material will be presented in class and/or provided online. Relevant lecture notes, papers, tutorials, and other resources will be posted on the course website.

Homeworks:

There are 4 homework sets. We will manage homework by email. Discussion of homework problems is permitted; however, each student must write and submit independent solutions.

Extensions will be granted with instructor approval in advance. Otherwise, late homework without such prior approval will not be accepted.

In-Class Presentation and Written Research Report:

Each student will give an in-class presentation and submit a written research report. The presentation accounts for 20% of the final grade, and the written research report accounts for 40% of the final grade. Detailed guidelines for the presentation and final project will be posted in the Resources section of the course website.

Policy on Re-grades:

Re-grades will be considered if you believe there is an error in the grading of your homework, presentation, or written report. You should explain the issue in writing and submit a detailed explanation to the instructor. A re-grade involves reconsideration of the relevant work and could result in a higher or lower grade than before.

Accommodations for Students with Disabilities:

If you think you need an accommodation for a disability, please let me know at your earliest convenience. Some aspects of this course, the assignments, the in-class activities, and the way the course is usually taught may be modified to facilitate your participation and progress.

Academic Integrity:

All students are subject to the University's academic integrity policies. A quick reference guide, as well as links to the official student code, can be found at:

<https://provost.illinois.edu/policies/policies/academic-integrity/students-quick-reference-guide-to-academ>